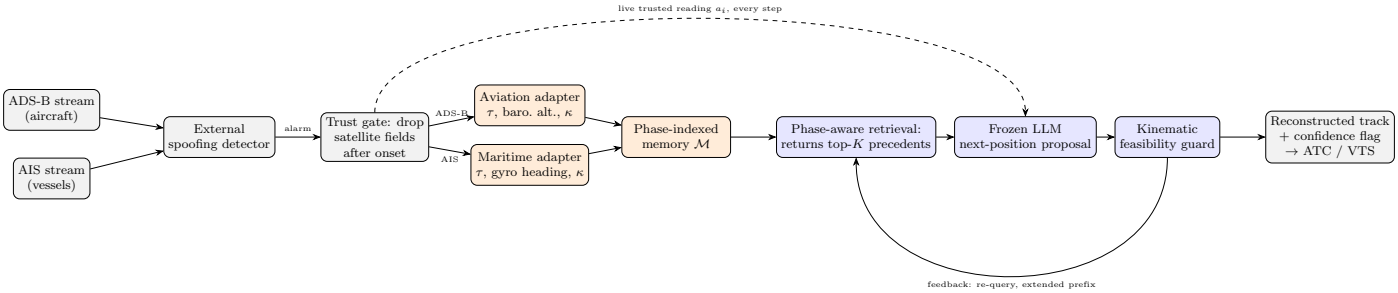
A Domain-Agnostic Reconstruction Pipeline for GNSS Spoofing

Method Overview: Aviation & Maritime

When an aircraft or vessel's satellite-navigation signal is spoofed, surveillance operators must discard its broadcast position yet keep tracking it. Existing trajectory predictors are trained for a single transport domain and must be rebuilt for each new one. **TrajRAG factors the problem into a shared, training-free engine and a thin, domain-specific adapter**, so that porting it from aircraft to ships requires no model training at all – only a prompt template, a memory index, a trusted non-satellite signal, and a short calibration run.

Shared Engine + Domain Adapter



Both surveillance streams share one detector, one trust gate, one retrieval engine, one frozen LLM, and one kinematic guard. Only the orange domain adapter (prompt template, trusted signal, kinematic constants, and the memory built from that domain's history) changes between aircraft and ships.

Aviation Data vs. Maritime Data

	Aviation	**Maritime**
Source	FlightRadar24 (ADS-B)	NOAA Marine Cadastre (AIS, public)
Corridors	Paris→Rome; Barcelona→Munich	Houston Ship Channel; Los Angeles/Long Beach approach
Trajectories	1,544 flights	3,506 commercial vessel voyages
Native sampling	≈120 s	Irregular; resampled to 300 s
Trusted signal	Barometric altitude (air-data system)	Gyrocompass true heading
Phases	Takeoff / cruise / landing	Departure / transit / arrival
Environmental context	Meteomatics (wind, turbulence)	Open-Meteo Marine (waves, swell, current)
LLM backbone	gpt-5.1, K=5 retrieved precedents	gpt-4.1, K=6 retrieved precedents

The adapter is the only thing that changes. The engine, the retrieval mechanism, the feedback loop, and the guard's functional form are byte-identical across both domains; only the five adapter elements above differ, and the kinematic tolerance constant is re-derived per domain from real data rather than reused (the one thing that did *not* transfer automatically between domains).

Maritime Results

Houston Ship Channel & Los Angeles/Long Beach

TrajRAG reconstructs a vessel's trajectory during a satellite-navigation blackout using a *frozen* large language model, retrieval of similar historical voyages, and a kinematic feasibility guard — with **no model training on maritime data**. We compare it against classical filters, recurrent networks, retrieval-only baselines, modern time-series forecasters, and AIS-ACNet, a convolutional network trained specifically on this task and corridor. Mean Haversine error (km, lower is better).

Main Result: Houston Ship Channel

Method	Departure	Transit	Arrival
Kalman filter	3.16	2.61	4.23
ARIMA	3.64	3.78	2.24
LSTM	10.44	7.99	6.42
BiLSTM	10.03	14.10	16.43
Mean-Retrieval (no LLM)	13.89	8.76	7.03
Numeric-kNN	5.53	3.93	3.17
AIS-ACNet (trained on this corridor)	**1.80**	**1.66**	**1.18**
TrajRAG (no training)	2.19	1.95	1.50

Also evaluated: seven modern forecasters (DLinear, PatchTST, iTransformer, Informer, Autoformer, FEDformer, TimesNet) on a shorter, exact-fit horizon – not directly comparable to the full-horizon methods above; TrajRAG and AIS-ACNet beat all seven on that same matched horizon.

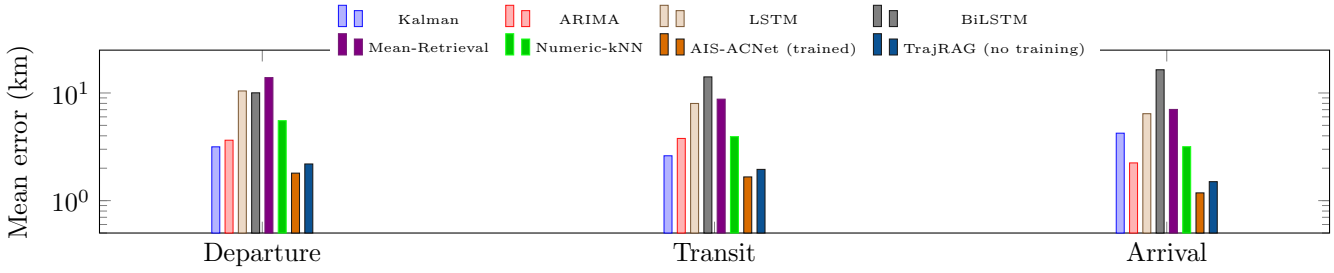
TrajRAG is statistically indistinguishable from AIS-ACNet in every phase (paired Wilcoxon test, $p = 0.77$ / 0.56 / 0.50) despite using no maritime training data at all. **TrajRAG significantly outperforms every classical, recurrent, and retrieval-only baseline in every phase, in both corridors** ($p \leq 6.5 \times 10^{-3}$, mostly $p < 10^{-4}$) — a 1.3–1.5× error reduction over the best classical method.

Ablation Study (Houston, matched pairs)

Configuration	Departure		Transit		Arrival	
	Mean	p	Mean	p	Mean	p
Full pipeline	2.56	–	1.63	–	1.47	–
w/o retrieval	2.59	0.93	2.42	**0.012**	2.20	**0.048**
w/o kinematic guard	3.23	0.20	1.59	0.28	1.97	0.76
w/o live heading signal	2.60	0.31	1.47	0.90	1.78	0.62

Retrieval is the dominant driver of accuracy: removing it significantly increases error in transit and arrival. **Removing the live trusted heading signal has no significant effect in any phase** — retrieval and the guard, not the live signal alone, appear to be the pipeline's main contributors.

Houston Ship Channel at a Glance



Caveat: results above follow a mid-project correction to two preprocessing/retrieval bugs; full-scale re-validation across the complete test corpus is ongoing future work.

Aviation Results

Paris→Rome & Barcelona→Munich

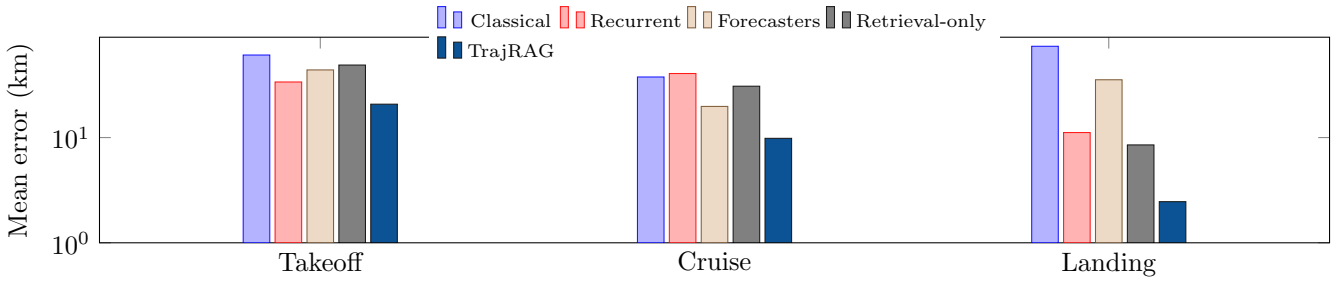
The aviation instantiation is the original TrajRAG configuration (conference version), reusing the identical shared engine that the maritime results on page 2 use. Mean Haversine error (km, lower is better) by flight phase, held-out test flights.

Main Result: Best Baseline per Family

Method family (best member)	Takeoff	Cruise	Landing
Classical: Kalman / ARIMA	60.95	37.69	73.82
Recurrent: LSTM / BiLSTM	33.78	40.66	11.17
Modern forecasters: DLinear / PatchTST / iTransformer	43.99	19.79	35.47
Retrieval only: Mean-Retrieval / Numeric-kNN	48.94	30.80	8.52
TrajRAG (gpt-5.1)	**20.78**	**9.84**	**2.46**

TrajRAG reduces mean error relative to the best member of every baseline family, in every flight phase — the largest gap is in landing (2.46 km vs. 8.52 km for the best retrieval-only method, a 3.5× reduction), the phase where a short, manoeuvre-dominated trajectory makes retrieved precedents most valuable. On four real interference episodes with independent multilateration ground truth, reconstructions stayed within 8 km of the reference position on average.

Aviation Ablation (conference study, qualitative summary)



Four component-level findings from the conference study's ablations carry over directly to the shared engine used here (numeric per-component table: conference paper, not reproduced in the journal extension):

- **Removing retrieval caused the largest accuracy degradation in every phase** — the same component that maritime's own ablation (page 2) identifies as its dominant contributor.
- **Replacing the LLM with plain neighbour averaging was significantly worse** — confirms the gain is not retrieval alone, mirroring the maritime Mean-Retrieval comparison.
- **The kinematic feasibility guard mainly reduced variance and worst-case error**, with a smaller effect on the mean — unlike maritime, where the guard also moves the mean in at least one phase (a genuine cross-domain difference).
- **Mixing several routes into one shared memory changed mean error by less than 1 km** — memories generalise across routes without a meaningful accuracy cost, supporting the one-service-many-corridors deployment model.